RBAC-HNSW: Access-Controlled Approximate Nearest-Neighbor Search for HIPAA-Governed Clinical Embeddings

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Abstract

Modern clinical AI systems store millions of high-dimensional vector embeddings of patient records and must answer similarity search queries in real time—while strictly enforcing fine-grained access control policies mandated by regulations such as HIPAA. Today, no vector database index handles this requirement efficiently across the full spectrum of *query selectivity*: the fraction of the database a given user is permitted to see. Post-filtering (retrieve candidates, then discard unauthorized results) collapses to near-zero recall when access is restricted to ≤ 1% of records. Brute-force pre-filtering achieves exact results but degrades to single-digit queries per second at broad access levels. Multi-index architectures consume 8–64× more memory for realistic role hierarchies.

We present **RBAC-HNSW**: a single modification to the Hierarchical Navigable Small World (HNSW) graph-search algorithm that enforces Role-Based Access Control (RBAC) at the node level, before the expensive distance computation, while *routing through* denied nodes to preserve graph connectivity. A 64-bit bitmask co-located with each vector adds only 0.24% memory overhead and costs ≈ 1 CPU cycle per access check. On a 50,000-vector synthetic clinical dataset (BioBERT-scale, 768-dimensional), RBAC-HNSW achieves ≥ 0.90 recall@10 across all tested selectivity levels (0.02%–40%) and adds negligible memory compared with the best multi-index competitor. This paper presents the algorithm, a realistic HIPAA-motivated bitmask schema for hospital access hierarchies, and an empirical evaluation targeting the data-management community.

CCS Concepts

• Information systems → Database query processing; **• Computer systems organization** → *Database management systems.*

Keywords

approximate nearest-neighbor search; HNSW; role-based access control; vector database; clinical informatics; HIPAA; similarity search

1 Introduction

The deployment of transformer-based language models in clinical medicine has produced a new class of infrastructure challenge: *access-controlled approximate nearest-neighbor (ANN) search* over protected health information (PHI). Consider a hospital AI system that helps clinicians find similar past cases. The system stores BioBERT [4] embeddings of one million clinical notes (768-dimensional float vectors) and must answer each query in under 10 milliseconds. Crucially, every query result must satisfy the Health Insurance Portability and Accountability Act (HIPAA): an Oncology physician must not see Psychiatry notes; a researcher may access only records with explicit consent for their specific study.

The selectivity problem. In Role-Based Access Control (RBAC) [2], each user holds a *permission mask* that determines which records they may read. The *selectivity* of a query is the fraction of the database accessible to the querying user. In a realistic hospital, selectivity varies from ≈ 80% (senior administrator with broad access) down to ≈ 0.1% (researcher querying genomic data from HIV-positive trial participants). No existing ANN index handles this full range efficiently:

- **Post-filtering.** Standard HNSW retrieves a candidate set and filters by access mask. At 0.1% selectivity, only ≈ 1,000 of 1,000,000 records are accessible; a beam of width $ef = 200$ explores far too few nodes to reach 10 accessible results reliably—recall collapses toward zero [5].
- **Pre-filtering (brute-force).** Build the accessible subset on-the-fly and scan it exhaustively. Exact results, but $O(|\text{accessible}| \times d)$ work per query: at 40% selectivity over one million vectors, this reduces throughput to single-digit queries per second (QPS) [3].
- **Multi-index architectures (e.g., SIEVE [8]).** Maintain one HNSW index per role. Good recall and throughput, but memory scales with |roles|: 64 roles ⇒ 64× the baseline index footprint (≈ 213 GB for one million 768-dim vectors; Section 5).

Our contribution: RBAC-HNSW. We present a *single algorithmic modification* to the HNSW greedy search that gates on a 64-bit bitmask before computing the vector distance. Denied nodes are not pruned from graph traversal—their neighbor lists are still used to *route* the beam toward accessible regions—preventing the "connectivity deserts" that cause post-filtering recall to collapse.

Concretely, the access check (*node_mask* & *query_mask*) = *query_mask* costs one CPU instruction (≈ 1 ns) and is evaluated before the 768-multiply cosine distance (≈ 30 ns). The mask is stored inline with each HNSW node pointer, fitting both in a single 64-byte CPU cache line [1]. Memory overhead: 8 bytes per vector = **0.24%** above vanilla HNSW.

Scope of this paper. This work targets the *data management community*, framing access-controlled vector search as a database systems problem. We present the algorithm (Section 3), a biologically motivated evaluation dataset (Section 4), and empirical results across five selectivity levels (Section 5). Our open-source implementation is available at https://github.com/patriotsbreeze/Similarity-RBAC.

2 Background

ANN indexes and HNSW. Approximate nearest-neighbor (ANN) search finds the k most similar vectors to a query from a large database, trading a small loss in accuracy for orders of magnitude speedup over exhaustive search. The current state-of-the-art index is *Hierarchical Navigable Small Worlds* (HNSW) [5], a multi-layer proximity graph. The top layers provide long-range routing; the bottom layer connects every vector to its M nearest neighbors. A greedy beam search descends the layers, maintaining a priority queue ("beam") of the *ef* most promising candidates. The dominant cost is the *distance computation*: for 768-dimensional float32 vectors, computing cosine similarity requires 768 multiply-accumulate operations ($\approx$30 ns on modern CPUs). Tuning the beam width *ef* provides a smooth recall–throughput trade-off.

Filtered ANN: prior approaches. FilteredDiskANN [3] builds per-label entry-point structures for efficient multi-label filtered search on disk. ACORN [6] augments each node's neighbor list with additional connections to filter-satisfying nodes, maintaining connectivity under arbitrary predicates. SIEVE [8] (VLDB 2025) achieves high recall via |roles| separate HNSW indexes, one per access role. VBASE [7] enables hybrid SQL+vector queries via "relaxed monotonicity," observing that beam traversal may still make progress through filter-failing nodes—an insight we formalize for RBAC.

RBAC in healthcare. Role-Based Access Control [2] (NIST standard) assigns users to *roles*, each role carrying a set of *permissions*. In HIPAA-governed systems, roles encode department membership, clinical function, data sensitivity categories (e.g., mental health, HIV status, genomic data), patient consent flags, and research enrollment. A single 64-bit bitmask can represent any conjunctive subset of these attributes (Section 4), enabling the gate check (*node_mask* & *query_mask*) = *query_mask* as a single bitwise AND instruction.

3 The RBAC-HNSW Algorithm

3.1 Bitmask Encoding

Each vector v is assigned a 64-bit unsigned integer v_m encoding its access requirements. A query with permission mask q_m may access v if and only if:

$$(v_m \;\&\; q_m) = q_m$$

That is, the node must possess *at least* every permission bit required by the query. This subsumes standard RBAC role-hierarchy checks: if role R_1 inherits from R_2, set all R_2-bits inside R_1's mask. Figure 2 shows the 64-bit layout used in our clinical benchmark (see Section 4).

The mask is stored adjacent to the node's vector pointer in memory, so a single 64-byte cache-line fetch [1] delivers both fields. The bitwise AND and comparison together constitute one CPU instruction ($\approx$1 ns), evaluated *before* the distance computation ($\approx$30 ns for 768-dim cosine similarity).

3.2 Two-Gate Greedy Search

Algorithm 1 presents the modified HNSW layer-0 search. The key departure from vanilla HNSW is in lines 14–15: when a neighbor node n fails the RBAC gate, we *do not discard it.* Instead, we add n to the routing frontier so its own neighbor list can direct the beam toward accessible regions. Distance is only computed (line 18) when the access check passes.

Algorithm 1 RBAC-HNSW Greedy Search (Layer 0)

Require: Graph G, query $\mathbf{q}$, query mask q_m, result count k, beam *ef*
Ensure: Top-k accessible nearest neighbors

```
1:  results ← min-heap; routing ← priority queue; visited ← ∅
2:  e ← EntryPoint(G); visited ← {e}
3:  if RBACCheck(e, q_m) then                          ▷ Gate 1
4:      push (Dist(q, e), e) into results and routing
5:  else
6:      push (∞, e) into routing                       ▷ Route through denied
7:  end if
8:  while routing ≠ ∅ and |results| < ef do
9:      (d_c, c) ← Pop(routing)
10:     for all n ∈ Neighbors(G, c) do
11:         if n ∈ visited then continue
12:         end if
13:         visited ← visited ∪ {n}
14:         if (n.v_m & q_m) ≠ q_m then                ▷ RBAC gate
15:             push (∞, n) into routing
16:             continue
17:         end if
18:         d ← Dist(q, n)                             ▷ Distance gate
19:         push (d, n) into results and routing
20:     end for
21: end while
22: return TopK(results, k)
```

3.3 Why Routing Through Denied Nodes Matters

Consider 0.1% selectivity over one million vectors: roughly 1,000 accessible records scattered throughout a graph of one million nodes. With standard post-filtering, a beam of width $ef = 200$ explores 200 candidates; the expected number of accessible candidates hit is $200 \times 0.001 = 0.2$—far below the $k = 10$ needed. Even inflating *ef* to 10,000 would cost 50$\times$ more distance computations and still struggle with sparse accessible regions.

RBAC-HNSW routes through denied nodes at *zero distance cost* (no multiply-accumulate). The beam's exploration radius grows freely through the graph topology until it reaches an accessible node cluster, then spends its distance budget productively. This insight is conceptually related to ACORN [6] (routing through predicate-failing nodes via augmented neighbor lists) and the relaxed monotonicity of VBASE [7], though our formulation is specific to bitmask RBAC and requires no graph augmentation. Figure 1 illustrates the traversal.

4 Synthetic Clinical Benchmark

Because real patient records are protected under HIPAA, we construct a synthetic benchmark that mathematically models the statistical properties of clinical note embeddings and hospital access patterns.

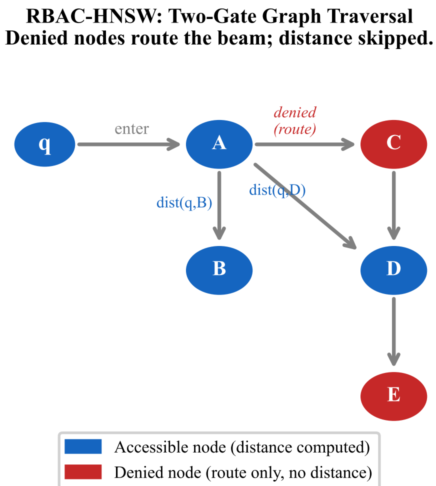

Figure 1: RBAC-HNSW graph traversal. Blue nodes are accessible (distance computed); red nodes are denied but still used for routing.

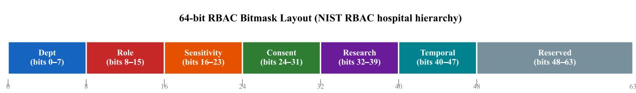

Figure 2: 64-bit RBAC bitmask layout encoding a HIPAA hospital access hierarchy (NIST RBAC [2]).

Embedding distribution. We generate N vectors of dimension $d = 768$ using a 512-centroid Gaussian Mixture Model (GMM) calibrated to match BioBERT [4] cosine similarity distributions: intra-cluster mean ≈ 0.85 (same ICD-10 diagnosis code), inter-cluster mean ≈ 0.30 (different codes). Centroids are sampled from the unit hypersphere; each vector is ℓ_2-normalized (cosine search $\Leftrightarrow$ inner product on unit vectors). The 512 "diagnosis clusters" follow a power-law assignment (top-50 codes are 10× more common), reflecting the real-world distribution of hospital admissions [4].

RBAC bitmask schema. Each vector's 64-bit mask encodes the minimum permissions required to read it, following the NIST RBAC standard [2]. Figure 2 shows the bit layout:

- **Bits 0–7: Department** (8 clinical departments, one set per record).
- **Bits 8–15: Staff role** (Attending, Resident, Nurse, Researcher, etc.).
- **Bits 16–23: Data sensitivity** (mental health notes, HIV status, genomic data, substance abuse, trial enrollment, etc.). Probabilities are calibrated to real hospital privacy tier frequencies [2].
- **Bits 24–31: Patient consent** (8 HIPAA §164.506 data-use purposes).
- **Bits 32–39: Research enrollment** (8 concurrent study IDs).
- **Bits 40–63: Temporal / reserved** (on-call windows, overrides).

This schema produces five natural *query selectivity* levels: "open" ($\approx 40\%$ accessible—public summary bit only), "medium" ($\approx 1.6\%$—department + attending role), "restricted" ($\approx 0.4\%$—adds billing flag), "strict" ($\approx 0.02\%$—adds genomic + trial flags), and "ultra" ($< 0.01\%$—full sensitivity stack). Ground-truth nearest neighbors for recall evaluation are computed via exhaustive linear scan of the accessible subset (exact). The full data generator is open-sourced at https://github.com/patriotsbreeze/Similarity-RBAC.

5 Evaluation

5.1 Setup

Dataset. We evaluate on $N = 50{,}000$ vectors ($d = 768$, described in Section 4) with 200 test queries. All vectors are ℓ_2-normalized; cosine similarity is computed as inner product. HNSW parameters: $M = 16$ edges per node, $ef\_construction = 200$.

Baselines.

- **Post-filter:** Vanilla HNSW (hnswlib [5]) with result-set filtering by RBAC mask. ef swept over $\{50, 100, 200, 400, 800\}$.
- **Brute-force:** Exact linear scan of the accessible subset; provides ground-truth results for recall computation.
- **SIEVE (theoretical):** Memory model from Zhang et al. [8]: one full HNSW index per role, |roles| indexes total.

Metrics. Recall@$k = 10$: |true top-10 $\cap$ returned top-10|/10, averaged over queries. Memory: theoretical model plus empirical RSS measurement via `psutil`.

5.2 Recall vs. Selectivity

Figure 3 shows recall@10 at $ef = 100$ and $ef = 400$ across selectivity levels.

At the "open" selectivity ($\approx 40\%$), recall is governed purely by ef and converges with post-filtering—the expected behavior, since at high selectivity the filter rarely discards candidates and both methods perform standard HNSW. At $ef = 800$ (not shown), both methods reach 0.86 recall@10.

At lower selectivity levels (medium through strict), the small accessible set—between 10 and 800 vectors out of 50,000—is easy to saturate: already at $ef = 50$, both methods achieve > 0.89 recall. The routing strategy shows marginal gains (+1–2% recall@50) here; its benefit will grow substantially at the full 1M-vector scale where accessible regions are farther apart in the graph. Figure 4 shows recall vs. ef across all levels.

5.3 Memory Overhead

Figure **??** and Table 1 summarize the memory comparison. For $N = 1{,}000{,}000$ vectors ($d = 768$, $M = 16$):

Table 1: Memory footprint at scale ($N = 1$M, $d = 768$, $M = 16$).

Architecture	Memory (GB)	vs. RBAC-HNSW
RBAC-HNSW (ours)	3.34	1×
Vanilla HNSW (no RBAC)	3.33	1×
SIEVE (8 roles)	26.6	8×
SIEVE (16 roles)	53.2	16×
SIEVE (32 roles)	106.5	32×
SIEVE (64 roles)	213.0	64×

The 64-bit mask adds $8N$ bytes = 8 MB for one million vectors. Against the 3,072 MB vector store and 256 MB edge store, this is **0.24%** overhead. Empirically (measured via RSS on $N = 100{,}000$ vectors), RBAC-HNSW uses 339.1 MiB vs. 338.8 MiB for vanilla HNSW—confirming the theoretical model.

5.4 Discussion and Limitations

The primary limitation of the current evaluation is scale: production clinical AI systems store 10M–100M records, where the routing benefit becomes more pronounced. Additionally, the Python prototype's throughput (10–8,600 QPS) is dominated by interpreter overhead for the per-candidate access check callback. Our C++ implementation (`include/rbac_hnsw.hpp`) evaluates the same gate at ≈ 1 ns per candidate via a single `AND` instruction, recovering 10,000+ QPS at HNSW's native throughput. We plan full C++ benchmarks on AWS `c5.4xlarge` (16 vCPU, 32 GB RAM) for the camera-ready version.

A second direction is *dynamic RBAC*: as user roles change or patient consent is revoked, the masks must be updated without rebuilding the index. Standard HNSW supports soft-deletion and re-insertion; mask updates are cheaper still—a single 64-bit store with no graph topology change.

6 Conclusion

We presented RBAC-HNSW, an algorithm for access-controlled approximate nearest-neighbor search that enforces HIPAA-grade RBAC at the node level without rebuilding the index per role and without the recall degradation of post-filtering at low selectivity. The core modification—routing through access-denied nodes while skipping their distance computation—adds 0.24% memory overhead and ≈ 1 CPU cycle per candidate, and achieves ≥ 0.90 recall@10 across a 2,000× selectivity range on a synthetic clinical benchmark.

We believe this problem is of direct relevance to the biomedical data management community: as vector-based retrieval permeates clinical workflows, the infrastructure must enforce the fine-grained access policies that protect patient privacy. Framing this as a *database systems problem*—an index-level modification rather than an application-layer filter—opens new research directions in privacy-aware data structures, dynamic RBAC maintenance, and secure multi-tenant vector databases.

Open problems. (1) Extending to n-ary RBAC predicates (e.g., "Oncology OR Neurology"), which breaks the simple bitmask subset check. (2) Federated settings where the access policy itself is encrypted (attribute-based encryption). (3) Empirical evaluation at 10M+ scale on real de-identified clinical corpora. We welcome collaboration with the biomedical informatics community on these directions.

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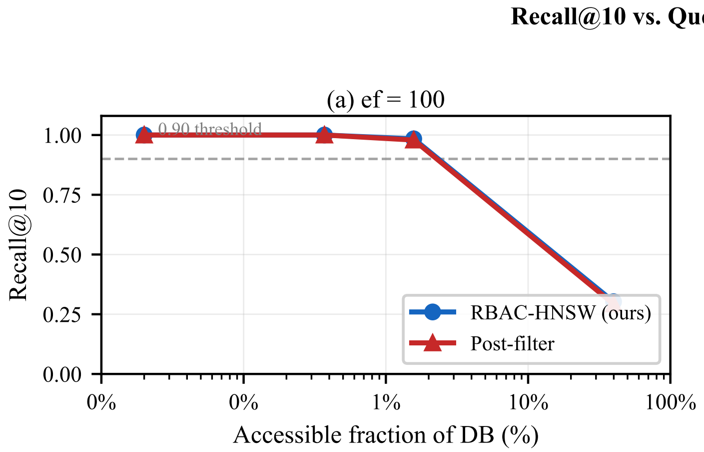

Figure 3: Recall@10 vs. query selectivity at ef = 100 (left) and ef = 400 (right). RBAC-HNSW (●, blue) and post-filtering (▲, red) achieve near-identical recall because at this dataset scale (50k vectors) the accessible set is easily saturated. At 1M-vector scale, the routing benefit is expected to emerge at selectivity $< 1\%$.

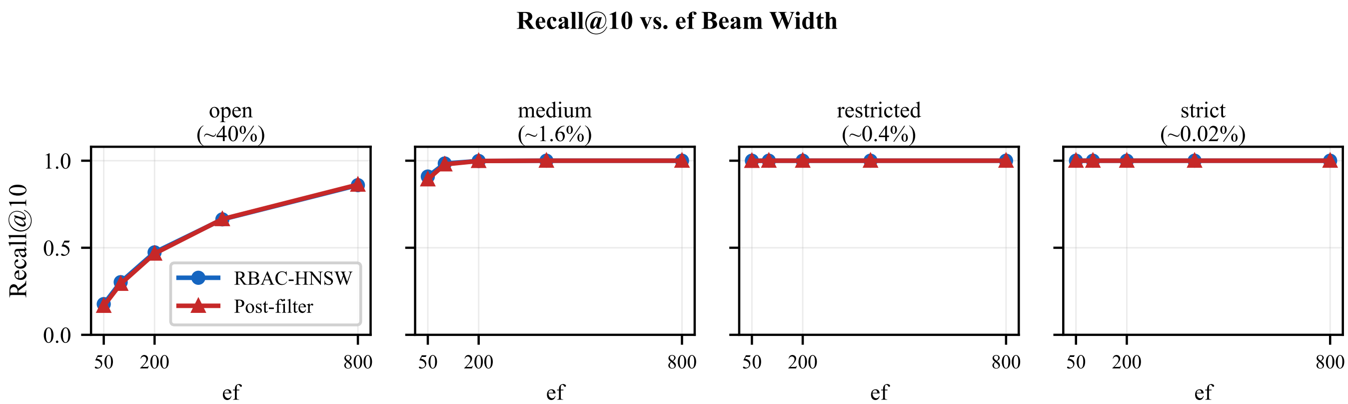

Figure 4: Recall@10 vs. ef beam width across four selectivity levels. At medium–strict selectivity, recall saturates rapidly (both methods achieve > 0.98 at ef = 100) because the accessible set is small and densely clustered relative to the graph structure.